

GrounUp Enterprise Specification

Phase 16 - GIS, Survey, Reality Capture & Drone Operations

Version 1.0.0 | Review-ready | Proposed lock: GES-P16-v1.0.0

1. Mission

Create a governed spatial operating system that unifies GIS, construction survey, reality capture and drone operations into accurate, traceable project information supporting estimating, operations, progress, quantities, documents and machine-control handoff.

2. Architecture Principles

CRS/datum/unit transparency; survey control integrity; source lineage; measurement QA/QC; versioned surfaces; large-spatial-data scalability; mobile/offline field collection; regulated-flight configurability; evidence-first AI; open interfaces.

3. Cross-Phase Integration

Phase 16 integrates the enterprise data model, estimating/earthwork quantities, project operations, AI construction intelligence, fleet assets, Phase 13 analytics, Phase 14 HSEQ, Phase 15 document control, and provides governed model/control handoff into Phase 17 machine control.

4. Survey & Coordinate Governance

Every survey, surface, raster, point cloud and derived quantity records its horizontal CRS/datum, vertical datum/geoid, units, source observations and processing lineage. Transformations are versioned and reviewable.

5. Drone Operations

Drone operations include aircraft, pilot/qualification, mission, flight plan, payload, battery, preflight/postflight, site/airspace authorization evidence, maintenance, media and incidents. Regulatory requirements are configurable rather than assumed static.

6. Reality Capture

Photogrammetry and LiDAR workflows manage calibration, ground control/checkpoints, processing runs, point clouds, orthomosaics, surfaces and acceptance evidence. Accuracy reports remain linked to the exact source and processing version.

7. Earthwork & Progress Intelligence

Surface comparisons support cut/fill, stockpiles, material movement, progress snapshots and change detection. Quantity outputs preserve units, assumptions such as shrink/swell where applicable, surface versions and evidence.

8. AI Geospatial Intelligence

AI can assist with feature recognition, object detection, terrain anomalies and change detection. Outputs retain evidence and confidence. AI cannot certify survey accuracy, approve regulated flight authority, or independently release machine-control data.

9. Machine Control Handoff

Phase 16 produces governed handoff packages containing approved surfaces/alignments, control/localization references, CRS/datum/units, checksums, revision and validation evidence for Phase 17.

Functional Domains

GIS Core

Enterprise geospatial layers, features, attributes, maps, symbology, topology, permissions and spatial search.

Coordinate Reference Systems

Horizontal/vertical datums, coordinate systems, units, transformations, geoid models and metadata.

Survey Control

Benchmarks, control points, monuments, baselines, traverses, networks, checks and adjustment evidence.

GNSS Survey

Base/rover workflows, RTK/RTN, static observations, corrections, epochs, precision and quality checks.

Total Station Survey

Instrument setup, backsight, foresight, traverses, stakeout, observations, prism constants and closures.

Digital Levels

Differential leveling, loops, benchmarks, closures, elevation transfer and field-book records.

Construction Layout

Points, lines, offsets, elevations, slopes, station/offset, stakeout, layout verification and as-staked records.

Topographic Survey

Topo points, breaklines, features, surfaces, contours, coding, checks and deliverables.

Existing Conditions

Utilities, structures, pavement, vegetation, drainage, terrain, site features and condition evidence.

Drone Operations

Aircraft, pilots, missions, batteries, payloads, airspace/authorization records, checklists, maintenance and flight logs.

Photogrammetry

Image capture, camera calibration, overlap, alignment, control, tie points, processing and accuracy.

LiDAR Reality Capture

Aerial/terrestrial/mobile LiDAR, trajectories, calibration, point clouds, classifications and accuracy.

Point Cloud Management

Point cloud storage, tiling, classification, filtering, registration, clipping, viewing and lineage.

Orthomosaic Management

Orthorectification, mosaics, resolution, tiles, control, publication and historical comparison.

Surface & Terrain Models

TINs, meshes, DEM/DSM/DTM, breaklines, contours, surface versions, comparisons and approvals.

Volumes & Earthwork

Cut/fill, stockpiles, excavation, embankment, borrow/waste, shrink/swell assumptions and quantity evidence.

Progress & Change Detection

Time-series surfaces/images, percent complete, elevation deltas, material movement and exception detection.

Corridor & Linear Infrastructure

Alignments, profiles, stationing, offsets, cross-sections, corridors, roads, utilities and linear assets.

Utility Mapping

Known/located utilities, potholes, locates, confidence, depth, survey evidence and clash references.

GIS Field Collection

Mobile forms, offline maps, GNSS capture, photos, sketches, inspections and synchronization.

Reality Capture QA/QC

Ground control, checkpoints, RMSE, residuals, density, completeness, calibration and acceptance.

Geofencing & Location Intelligence

Project limits, exclusion zones, work zones, haul routes, asset zones, alerts and spatial rules.

Drone Media & Inspection

Aerial photos/video, inspection missions, annotated observations, progress media and evidence.

Regulatory & Flight Compliance

Configurable aviation, privacy, site, owner and company requirements with documented approvals/evidence.

Survey Deliverables

Field books, point files, CAD/GIS exports, reports, plats, surfaces, certificates and transmittals.

Machine Control Handoff

Governed export interfaces for models, surfaces, alignments, localization/control and Phase 17 handoff.

Spatial Analytics

Heat maps, nearest/within analysis, spatial joins, route analysis, quantities, trends and geospatial KPIs.

AI Geospatial Intelligence

Evidence-grounded feature recognition, change detection, surface analysis, anomaly detection and recommendations.

Geospatial Digital Twin

Current spatial state linked to project, documents, quantities, assets, field work and historical reality capture.

Controlled Artifact Summary

Artifact	Count
Requirements	360
Entities	165
Fields	260
Business Rules	70
Validation Rules	70
Workflows	48
Engines	49
Api Resources	60
Ui Surfaces	42
Reports	34
Kpis	48
Tests	360

KPI Registry

P16-KPI-001 - Control Point Verification % - GIS Core - Governed metric definition for Control Point Verification %.

P16-KPI-002 - Survey Closure Pass Rate - Coordinate Reference Systems - Governed metric definition for Survey Closure Pass Rate.

P16-KPI-003 - GNSS Fix Quality % - Survey Control - Governed metric definition for GNSS Fix Quality %.

P16-KPI-004 - Horizontal Accuracy Pass Rate - GNSS Survey - Governed metric definition for Horizontal Accuracy Pass Rate.

P16-KPI-005 - Vertical Accuracy Pass Rate - Total Station Survey - Governed metric definition for Vertical Accuracy Pass Rate.

P16-KPI-006 - Stakeout Verification % - Digital Levels - Governed metric definition for Stakeout Verification %.

P16-KPI-007 - Survey Rework Rate - Construction Layout - Governed metric definition for Survey Rework Rate.

P16-KPI-008 - Topo Completeness - Topographic Survey - Governed metric definition for Topo Completeness.

P16-KPI-009 - Utility Confidence Coverage - Existing Conditions - Governed metric definition for Utility Confidence Coverage.

P16-KPI-010 - Drone Mission Completion - Drone Operations - Governed metric definition for Drone Mission Completion.

P16-KPI-011 - Flight Compliance % - Photogrammetry - Governed metric definition for Flight Compliance %.

P16-KPI-012 - Preflight Checklist Completion - LiDAR Reality Capture - Governed metric definition for Preflight Checklist Completion.

P16-KPI-013 - Battery Readiness % - Point Cloud Management - Governed metric definition for Battery Readiness %.

P16-KPI-014 - Mission Reflight Rate - Orthomosaic Management - Governed metric definition for Mission Reflight Rate.

P16-KPI-015 - Photogrammetry RMSE - Surface & Terrain Models - Governed metric definition for Photogrammetry RMSE.

P16-KPI-016 - LiDAR RMSE - Volumes & Earthwork - Governed metric definition for LiDAR RMSE.

P16-KPI-017 - Point Cloud Density Compliance - Progress & Change Detection - Governed metric definition for Point Cloud Density Compliance.

P16-KPI-018 - Point Cloud Classification QA - Corridor & Linear Infrastructure - Governed metric definition for Point Cloud Classification QA.

P16-KPI-019 - Orthomosaic GSD Compliance - Utility Mapping - Governed metric definition for Orthomosaic GSD Compliance.

P16-KPI-020 - Surface QA Pass Rate - GIS Field Collection - Governed metric definition for Surface QA Pass Rate.

P16-KPI-021 - Cut/Fill Quantity Confidence - Reality Capture QA/QC - Governed metric definition for Cut/Fill Quantity Confidence.

P16-KPI-022 - Stockpile Volume Accuracy - Geofencing & Location Intelligence - Governed metric definition for Stockpile Volume Accuracy.

P16-KPI-023 - Progress Capture Frequency - Drone Media & Inspection - Governed metric definition for Progress Capture Frequency.

P16-KPI-024 - Change Detection Review Closure - Regulatory & Flight Compliance - Governed metric definition for Change Detection Review Closure.

P16-KPI-025 - Field GIS Sync Success - Survey Deliverables - Governed metric definition for Field GIS Sync Success.

P16-KPI-026 - Offline Map Currency - Machine Control Handoff - Governed metric definition for Offline Map Currency.

P16-KPI-027 - Geofence Alert Closure - Spatial Analytics - Governed metric definition for Geofence Alert Closure.

P16-KPI-028 - Survey Deliverable On-Time % - AI Geospatial Intelligence - Governed metric definition for Survey Deliverable On-Time %.

P16-KPI-029 - Machine Control Handoff Pass Rate - Geospatial Digital Twin - Governed metric definition for Machine Control Handoff Pass Rate.

P16-KPI-030 - Coordinate Transformation Exceptions - GIS Core - Governed metric definition for Coordinate Transformation Exceptions.

P16-KPI-031 - Datum/Geoid Metadata Completeness - Coordinate Reference Systems - Governed metric definition for Datum/Geoid Metadata Completeness.

P16-KPI-032 - Spatial Data Quality Pass Rate - Survey Control - Governed metric definition for Spatial Data Quality Pass Rate.

P16-KPI-033 - Spatial Layer Freshness - GNSS Survey - Governed metric definition for Spatial Layer Freshness.

P16-KPI-034 - Control Network Freshness - Total Station Survey - Governed metric definition for Control Network Freshness.

P16-KPI-035 - Reality Capture Processing Cycle Time - Digital Levels - Governed metric definition for Reality Capture Processing Cycle Time.

P16-KPI-036 - Reality Capture QA Cycle Time - Construction Layout - Governed metric definition for Reality Capture QA Cycle Time.

P16-KPI-037 - Drone Utilization - Topographic Survey - Governed metric definition for Drone Utilization.

P16-KPI-038 - Pilot Qualification Compliance - Existing Conditions - Governed metric definition for Pilot Qualification Compliance.

P16-KPI-039 - Aircraft Maintenance Compliance - Drone Operations - Governed metric definition for Aircraft Maintenance Compliance.

P16-KPI-040 - Spatial Query Performance - Photogrammetry - Governed metric definition for Spatial Query Performance.

P16-KPI-041 - Map Adoption - LiDAR Reality Capture - Governed metric definition for Map Adoption.

P16-KPI-042 - Spatial Analytics Adoption - Point Cloud Management - Governed metric definition for Spatial Analytics Adoption.

P16-KPI-043 - AI Feature Detection Acceptance - Orthomosaic Management - Governed metric definition for AI Feature Detection Acceptance.

P16-KPI-044 - AI Change Detection Acceptance - Surface & Terrain Models - Governed metric definition for AI Change Detection Acceptance.

P16-KPI-045 - AI Evidence Coverage - Volumes & Earthwork - Governed metric definition for AI Evidence Coverage.

P16-KPI-046 - Geospatial Digital Twin Freshness - Progress & Change Detection - Governed metric definition for Geospatial Digital Twin Freshness.

P16-KPI-047 - Spatial Data Storage Growth - Corridor & Linear Infrastructure - Governed metric definition for Spatial Data Storage Growth.

P16-KPI-048 - Survey Cost per Project - Utility Mapping - Governed metric definition for Survey Cost per Project.

Engine Registry

P16-ENG-001 - Coordinate Transformation Engine - Deterministic - Provides coordinate transformation engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-002 - Datum/Geoid Resolution Engine - Deterministic - Provides datum/geoid resolution engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-003 - Geometry Validation Engine - Deterministic - Provides geometry validation engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-004 - Topology Validation Engine - Deterministic - Provides topology validation engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-005 - Spatial Index Engine - Deterministic - Provides spatial index engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-006 - Survey Control Engine - Deterministic - Provides survey control engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-007 - Network Adjustment Engine - Deterministic - Provides network adjustment engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-008 - Closure Calculation Engine - Deterministic - Provides closure calculation engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-009 - GNSS Quality Engine - Deterministic - Provides gnss quality engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-010 - RTK/RTN Validation Engine - Deterministic - Provides rtk/rtn validation engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-011 - Total Station Reduction Engine - Deterministic - Provides total station reduction engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-012 - Level Closure Engine - Deterministic - Provides level closure engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-013 - Stakeout Calculation Engine - Deterministic - Provides stakeout calculation engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-014 - Station/Offset Engine - Deterministic - Provides station/offset engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-015 - Feature Coding Engine - Deterministic - Provides feature coding engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-016 - Photogrammetry Alignment Engine - Deterministic - Provides photogrammetry alignment engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-017 - Ground Control Adjustment Engine - Deterministic - Provides ground control adjustment engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-018 - Orthomosaic Engine - Deterministic - Provides orthomosaic engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-019 - LiDAR Trajectory Engine - Deterministic - Provides lidar trajectory engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-020 - Point Cloud Registration Engine - Deterministic - Provides point cloud registration engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-021 - Point Classification Engine - Deterministic - Provides point classification engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-022 - Surface/TIN Engine - Deterministic - Provides surface/tin engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-023 - Contour Engine - Deterministic - Provides contour engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-024 - Cross-Section Engine - Deterministic - Provides cross-section engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-025 - Profile Engine - Deterministic - Provides profile engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-026 - Cut/Fill Engine - Deterministic - Provides cut/fill engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-027 - Stockpile Volume Engine - Deterministic - Provides stockpile volume engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-028 - Swell/Shrink Conversion Engine - Deterministic - Provides swell/shrink conversion engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-029 - Progress Comparison Engine - Deterministic - Provides progress comparison engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-030 - Change Detection Engine - Deterministic - Provides change detection engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-031 - Geofence Engine - Deterministic - Provides geofence engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-032 - Route/Proximity Engine - Deterministic - Provides route/proximity engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-033 - Spatial Join Engine - Deterministic - Provides spatial join engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-034 - Field Offline Sync Engine - Deterministic - Provides field offline sync engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-035 - Flight Planning Engine - Deterministic - Provides flight planning engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-036 - Flight Compliance Engine - Deterministic - Provides flight compliance engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-037 - Battery Readiness Engine - Deterministic - Provides battery readiness engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-038 - Payload Readiness Engine - Deterministic - Provides payload readiness engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-039 - Reality Capture QA Engine - Deterministic - Provides reality capture qa engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-040 - RMSE/Accuracy Engine - Deterministic - Provides rmse/accuracy engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-041 - Machine Control Handoff Validator - Deterministic - Provides machine control handoff validator using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-042 - Survey Deliverable Engine - Deterministic - Provides survey deliverable engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-043 - Spatial KPI Engine - Deterministic - Provides spatial kpi engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-044 - Feature Recognition Engine - AI-assisted - Provides feature recognition engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-045 - Terrain Anomaly Engine - AI-assisted - Provides terrain anomaly engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-046 - AI Change Detection Engine - AI-assisted - Provides ai change detection engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-047 - Evidence Retrieval Engine - AI-assisted - Provides evidence retrieval engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-048 - AI Confidence Engine - AI-assisted - Provides ai confidence engine using governed spatial inputs, reference systems, quality criteria and lineage.

P16-ENG-049 - Geospatial Digital Twin Engine - AI-assisted - Provides geospatial digital twin engine using governed spatial inputs, reference systems, quality criteria and lineage.

Workflow Registry

P16-WF-001 - GIS Layer Creation & Publication - GIS Core - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-002 - Coordinate System Assignment - Coordinate Reference Systems - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-003 - Coordinate Transformation Review - Survey Control - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-004 - Survey Control Establishment - GNSS Survey - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-005 - Control Network Adjustment - Total Station Survey - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-006 - Benchmark Verification - Digital Levels - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-007 - GNSS RTK Survey - Construction Layout - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-008 - GNSS Static Survey - Topographic Survey - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-009 - RTN/Correction Validation - Existing Conditions - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-010 - Total Station Setup & Traverse - Drone Operations - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-011 - Digital Level Loop - Photogrammetry - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-012 - Construction Stakeout - LiDAR Reality Capture - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-013 - Stakeout Verification - Point Cloud Management - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-014 - Topographic Survey - Orthomosaic Management - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-015 - Existing Conditions Capture - Surface & Terrain Models - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-016 - Utility Locate & Pothole Capture - Volumes & Earthwork - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-017 - Drone Mission Request - Progress & Change Detection - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-018 - Drone Mission Planning - Corridor & Linear Infrastructure - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-019 - Flight Compliance Review - Utility Mapping - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-020 - Aircraft/Payload Preflight - GIS Field Collection - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-021 - Drone Flight Execution - Reality Capture QA/QC - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-022 - Drone Postflight - Geofencing & Location Intelligence - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-023 - Battery Lifecycle Update - Drone Media & Inspection - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-024 - Photogrammetry Processing - Regulatory & Flight Compliance - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-025 - LiDAR Processing - Survey Deliverables - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-026 - Ground Control/Checkpoint QA - Machine Control Handoff - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-027 - Point Cloud Registration - Spatial Analytics - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-028 - Point Cloud Classification - AI Geospatial Intelligence - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-029 - Orthomosaic Publication - Geospatial Digital Twin - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-030 - Surface/Terrain Model Creation - GIS Core - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-031 - Surface QA & Approval - Coordinate Reference Systems - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-032 - Cut/Fill Volume Calculation - Survey Control - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-033 - Stockpile Volume Calculation - GNSS Survey - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-034 - Progress Capture & Comparison - Total Station Survey - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-035 - Change Detection Review - Digital Levels - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-036 - Corridor/Stationing Publication - Construction Layout - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-037 - Field GIS Collection - Topographic Survey - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-038 - Offline Map Sync - Existing Conditions - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-039 - Geofence/Spatial Alert Publication - Drone Operations - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-040 - Aerial Inspection Review - Photogrammetry - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-041 - Survey Deliverable Production - LiDAR Reality Capture - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-042 - Machine Control Handoff Package - Point Cloud Management - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-043 - Spatial Data Import - Orthomosaic Management - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-044 - Spatial Data Export - Surface & Terrain Models - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-045 - Geospatial Digital Twin Refresh - Volumes & Earthwork - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-046 - AI Feature/Change Detection Review - Progress & Change Detection - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-047 - Reality Capture Release & Rollback - Corridor & Linear Infrastructure - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

P16-WF-048 - Survey Audit & Closeout - Utility Mapping - Initiate > Validate > Capture/Process > QA/QC > Review > Approve > Publish/Export > Archive

Spatial Governance

All released spatial products preserve CRS/datum/unit metadata, source lineage and QA evidence. AI remains advisory for survey certification, regulated flight authority and machine-control release.